AgroNex-Pi: Design and Implementation of an Autonomous Smart Agricultural Robot with Edge AI Plant Disease Detection, GPS Navigation, and Cloud Telemetry

Nikhil [Author Name], [Co-Author 1], [Co-Author 2], [Co-Author 3]

Department of [Electronics/Computer Engineering], [Institute Name], [City, State, Country]

Email: [author@institute.edu]

Abstract

This paper presents AgroNex-Pi, a fully autonomous smart agricultural robot built on the Raspberry Pi 4 platform that integrates five functional subsystems: real-time soil moisture sensing with automated drip irrigation, scheduled grass trimming, GPS + compass guided waypoint navigation, edge AI plant disease detection using TensorFlow Lite MobileNetV2, and live cloud telemetry via Amazon Web Services (AWS). The plant disease detection subsystem employs a dual-model architecture capable of classifying five Tulsi (Ocimum tenuiflorum) disease states and eight PlantVillage crop classes (Tomato, Potato, Bell Pepper) with a minimum confidence threshold of 70% and a majority-voting image segmentation scheme to suppress false positives. All subsystems are coordinated by a multi-threaded Python supervisor implementing a Finite State Machine (FSM) with hardware interlocks to prevent simultaneous high-current actuator operation. A serverless AWS stack (API Gateway + Lambda + S3) provides remote monitoring and bidirectional control. Experimental results demonstrate [INSERT RESULTS: classification accuracy, irrigation response time, navigation precision]. The system achieves [INSERT PERFORMANCE METRICS] making it suitable for deployment in small-scale farms, home gardens, and nurseries.

Index Terms *— Autonomous Agricultural Robot, Edge AI, TensorFlow Lite, Plant Disease Detection, Raspberry Pi, IoT, GPS Navigation, Smart Irrigation, MobileNetV2, Transfer Learning.*

I. INTRODUCTION

Agriculture remains the backbone of economies in developing nations, yet it faces mounting challenges including labor shortages, water scarcity, and the rapid spread of plant diseases. According to the Food and Agriculture Organization (FAO), plant pathogens cause annual crop losses estimated at 10-40% of global food production [1]. Early detection and intervention are critical but traditionally require expert agronomists visiting fields regularly — a resource constraint that small-scale farmers cannot sustain.

The emergence of low-cost single-board computers (SBCs) such as the Raspberry Pi 4, combined with advances in lightweight neural network architectures like MobileNetV2 [2] and the TensorFlow Lite runtime, has made edge AI inference feasible on embedded systems with watt-level power budgets. Simultaneously, cloud platforms like AWS provide serverless infrastructure for remote monitoring without requiring dedicated server maintenance.

This work presents AgroNex-Pi, which integrates soil sensing, automated irrigation, grass trimming, GPS-guided navigation, and AI-powered plant disease detection into a single autonomous rover. The system targets Tulsi (Holy Basil, Ocimum tenuiflorum) as the primary disease detection subject — a plant of significant cultural and medicinal importance in South Asia — while also supporting broad-crop PlantVillage classification.

A. Contributions

The principal contributions of this work are:

1) A fully integrated autonomous agricultural rover combining five subsystems on the Raspberry Pi 4 platform.

2) A dual-model TFLite architecture with a switchable model profile supporting local plants (Tulsi) and broad agricultural crops.

3) A majority-voting image segmentation scheme requiring >50% of image segments to be diseased before alerting, reducing false positives.

4) A hardware interlock system preventing simultaneous high-current actuator operation (pump + cutter + motors).

5) A serverless AWS telemetry stack enabling remote monitoring and bidirectional control from any internet-connected device.

II. RELATED WORK

Automated agricultural robots have been explored extensively in precision agriculture research. Bac et al. [3] demonstrated robotic harvesting in greenhouse environments using visual servoing but required structured environments with known plant geometry. Mohanty et al. [4] showed that deep convolutional neural networks trained on the PlantVillage dataset could achieve greater than 99% accuracy in controlled laboratory conditions, though real-field performance remained lower due to background clutter and lighting variation.

Smartphone-based disease diagnosis apps (e.g., Plantix, PlantSnap) deploy cloud inference models requiring internet connectivity for every prediction. In contrast, edge deployment on-device eliminates latency and operates offline. Howard et al. [2] introduced MobileNetV2, a depthwise separable convolution architecture optimized for mobile and embedded deployment, achieving ImageNet accuracy competitive with VGG-16 at a fraction of the computational cost.

IoT-based irrigation systems [5] have demonstrated 30-40% water savings over timer-based systems, while GPS-guided agricultural robots [6] have shown promise for field navigation. However, prior work has not integrated all five subsystems into a single embedded platform at low cost. AgroNex-Pi addresses this gap.

III. SYSTEM DESIGN AND ARCHITECTURE

AgroNex-Pi follows a layered software architecture with strict separation of concerns across nine Python modules. The hardware abstraction layer (HAL) in hardware.py encapsulates all GPIO, SPI, I2C, and camera access, ensuring that no other module accesses physical registers directly.

A. Hardware Platform

The rover is built around the Raspberry Pi 4 Model B (4 GB RAM, 64-bit quad-core Cortex-A72 at 1.5 GHz). Analog sensor inputs are digitized by an ADS1115 16-bit I2C ADC, with an MCP3008 SPI ADC available as fallback. Motor drive is provided by dual BTS7960 43A H-bridge modules controlled via software PWM at 1 kHz. Table I lists the complete bill of materials.

TABLE I - HARDWARE COMPONENTS

Component	Model / Specification	Purpose
SBC	Raspberry Pi 4B (4GB)	Central processor
ADC (I2C)	ADS1115 16-bit	Soil + Water sensors
Soil Moisture	Capacitive v1.2 (3.3V)	Moisture sensing
Water Level	HW-038 Analog 3-pin	Reservoir monitoring
Temp/Humidity	DHT11/22	Ambient conditions

GPS	NEO-7M UART	Positioning
Compass	HMC5883L I2C	Heading feedback
Camera	Pi Cam v2 / USB webcam	Leaf image capture
Motor Driver	BTS7960 43A (x2)	Drive motor control
Relay Module	2-ch 5V Active LOW	Pump + cutter switch
Battery	12V 7Ah SLA	Power supply

B. Software Architecture

The system supervisor (main.py) implements a Finite State Machine with states: BOOT, SELF_TEST, IDLE, IRRIGATING, CUTTING, NAVIGATING, SCANNING, and SAFE_STATE. Four concurrent threads are managed: the 3-second Control Loop, the Disease Scanner, the Flask REST API Server (port 5000), and the Watchdog Timer. Fig. 1 illustrates the software architecture.

Fig. 1. AgroNex-Pi multi-threaded software architecture and FSM state diagram.

C. Hardware Wiring and GPIO Assignment

All GPIO pins use BCM numbering. Signal pins: DHT data (GPIO 4), Relay pump (GPIO 17, Active LOW), Relay cutter (GPIO 27, Active LOW), Buzzer (GPIO 22), LED (GPIO 23). Motor PWM: Left RPWM/LPWM (GPIO 5/6), Right RPWM/LPWM (GPIO 13/19). I2C bus (GPIO 2=SDA, GPIO 3=SCL) carries ADS1115 (0x48) and HMC5883L (0x1E). GPS NEO-7M connects to /dev/serial0 (UART RX = GPIO 15).

IV. SUBSYSTEM IMPLEMENTATION

A. Soil Moisture Sensing and Automated Irrigation

A capacitive soil moisture sensor (v1.2) is connected to ADS1115 channel A0. Raw 16-bit ADC counts are linearized to moisture percentage using two-point calibration: ADS_SOIL_DRY = 21000 (0% in dry air) and ADS_SOIL_WET = 9500 (100% submerged). A hysteresis band governs pump activation: pump starts when moisture drops below 35% and stops when it reaches 65% or after a 60-second safety timeout. A mandatory 120-second cooldown protects relay contacts.

B. GPS Waypoint Navigation

Navigation employs a NEO-7M GPS module (UART, 9600 baud) and an HMC5883L 3-axis magnetometer. The Haversine formula computes great-circle distance and bearing to the target waypoint. The rover executes motion in brief bursts (1.0s forward, 0.2s pivot) and re-samples sensors between each burst. A heading error tolerance of plus or minus 12 degrees determines whether to drive forward or pivot. Waypoint arrival is declared when distance is less than or equal to 2.5 m.

C. Plant Disease Detection

The disease detection pipeline begins with a green-pixel pre-filter: the HSV-transformed camera frame must contain at least 15% green pixels (H: 35-90 degrees, S: 40-255, V: 40-255) before inference is triggered. An image quality gate rejects frames with Laplacian variance below 12.0 (blur) or luminance outside the range 15-250 (poor exposure).

The accepted frame is resized to 224x224 and normalized to [0.0, 1.0] float32. To reduce false positives from background pixels, the image is divided into a 3x3 grid of nine segments. Each segment is classified independently by the TFLite MobileNetV2 model. A disease alert is issued only if more than 50% of segments (more than 4 of 9) are classified

as diseased with confidence at least 70%. This majority-voting scheme is illustrated in Fig. 2.

[Fig. 2: 3x3 Image Segmentation Majority Voting Scheme]

(Insert figure / photograph here)

Fig. 2. Majority-voting segmentation: disease declared only when more than 50% of segments are classified as diseased.

D. Dual-Model Architecture

A model profile switch (AGRONEX_MODEL_PROFILE environment variable) selects between two TFLite models at startup: (1) the Custom Tulsi Model trained on locally collected images (5 classes) and (2) the PlantVillage Model trained on the publicly available PlantVillage dataset (8 classes). The TFLite runtime is loaded via a three-level fallback chain supporting tflite_runtime, ai_edge_litert, and full TensorFlow.

V. AI MODEL TRAINING METHODOLOGY

A. Dataset Collection

The custom Tulsi disease dataset was collected using the Raspberry Pi Camera v2 under natural outdoor lighting conditions. [INSERT NUMBER] images per class were captured across four disease states and one background class. The PlantVillage dataset [4] was used for broad-crop training, containing over 54,000 images across 38 plant-disease combinations, of which 8 target classes were selected.

B. Model Architecture and Training

Transfer learning from MobileNetV2 (pretrained on ImageNet, input 224x224x3) was employed. The base model was frozen for initial training, followed by fine-tuning of the last 50 layers. A custom classification head was added: GlobalAveragePooling2D, Dropout(0.4), Dense(128, ReLU), Dropout(0.3), Dense(N_classes, Softmax). Training used the Adam optimizer with categorical cross-entropy loss. Data augmentation included random horizontal flip, rotation of plus or minus 15 degrees, zoom of plus or minus 20%, and brightness variation of plus or minus 30%.

TABLE II - MODEL TRAINING PARAMETERS

Parameter	Custom Tulsi Model	PlantVillage Model
Base Architecture	MobileNetV2	MobileNetV2
Input Resolution	224 x 224 x 3	224 x 224 x 3
Classes	5	8
Initial Learning Rate	1e-4	1e-4
Fine-tune Learning Rate	1e-5	1e-5
Initial Epochs	15	20
Fine-tune Epochs	10	10
Batch Size	32	32
Optimizer	Adam	Adam
Export Format	TFLite float32	TFLite float32

VI. CLOUD ARCHITECTURE AND REMOTE MONITORING

A serverless AWS stack enables remote monitoring without dedicated server infrastructure. The Raspberry Pi periodically (every 5 seconds) HTTP POSTs sensor telemetry JSON to AWS API Gateway, which invokes a Python Lambda function that writes the payload to an Amazon S3 bucket as state.json. The S3-hosted dashboard frontend fetches this file at 3-second intervals to display live telemetry. Disease scan JPEG images are uploaded asynchronously after each inference. A bidirectional command channel allows the cloud dashboard to remotely trigger irrigation, scanning, and navigation commands. Fig. 3 illustrates the AWS data flow.

[Fig. 3: AWS Cloud Architecture Data Flow Diagram]

(Insert figure / photograph here)

Fig. 3. AWS serverless architecture: Pi to API Gateway to Lambda to S3, with S3-hosted dashboard polling.

VII. EXPERIMENTAL RESULTS AND EVALUATION

This section presents experimental results for each subsystem. All experiments were conducted on the physical rover hardware under [INSERT CONDITIONS: indoor/outdoor, lighting conditions, test location].

A. Disease Detection Performance

[Fig. 4: Confusion Matrix — Tulsi Disease Classification Results]

(Insert figure / photograph here)

Fig. 4. Confusion matrix for Tulsi disease classification on the test set.

TABLE III - DISEASE DETECTION CLASSIFICATION METRICS

Class	Precision	Recall	F1-Score	Support
Tulsi___Healthy	[X.XX]	[X.XX]	[X.XX]	[N]
Tulsi___Infected	[X.XX]	[X.XX]	[X.XX]	[N]
Tulsi___Infested	[X.XX]	[X.XX]	[X.XX]	[N]
Tulsi___Wilting	[X.XX]	[X.XX]	[X.XX]	[N]
Background	[X.XX]	[X.XX]	[X.XX]	[N]
Weighted Average	[X.XX]	[X.XX]	[X.XX]	[N]

B. Irrigation System Response

[Fig. 5: Soil Moisture vs. Time — Irrigation Cycle Response Chart]

(Insert figure / photograph here)

Fig. 5. Soil moisture percentage over time showing irrigation trigger at 35% and pump-off at 65%.

The hysteresis controller maintained soil moisture between 35%-65% across [N] irrigation cycles during testing. Average irrigation duration was [X] plus or minus [Y] seconds. No false pump timeouts (PUMP_SAFETY_TIMEOUT) were recorded during normal operation.

C. Navigation Accuracy

[Fig. 6: GPS Navigation Path — Planned vs. Actual Route]

(Insert figure / photograph here)

Fig. 6. Planned navigation route (blue) vs. actual rover GPS track (red) for a [N]-waypoint mission.

TABLE IV - NAVIGATION PERFORMANCE METRICS

Metric	Value
Number of waypoints tested	[N]
Waypoint arrival accuracy (mean error)	[X.X] m
Heading alignment accuracy (mean error)	[X.X] degrees
Average time per waypoint	[X] seconds
GPS fix availability	[X]% uptime

D. System Resource Utilization on Raspberry Pi 4

TABLE V - RASPBERRY PI 4 RESOURCE USAGE

Metric	Idle (No Inference)	During TFLite Inference
CPU Usage	[X]%	[X]%
RAM Usage	[X] MB	[X] MB
Inference Latency	—	[X] ms per frame
Power Consumption	[X] W	[X] W
CPU Temperature	[X] deg C	[X] deg C

E. System Photographs

[Fig. 7: AgroNex-Pi Rover — Physical Hardware Assembly Photo]

(Insert figure / photograph here)

Fig. 7. AgroNex-Pi physical rover showing chassis, sensors, motor drivers, and Raspberry Pi 4.

[Fig. 8: AgroNex-Pi Web Dashboard — Live Telemetry Screenshot]

(Insert figure / photograph here)

Fig. 8. AWS S3-hosted web dashboard displaying live sensor telemetry and disease scan results.

VIII. DISCUSSION

The experimental results confirm that MobileNetV2-based transfer learning on locally collected Tulsi disease images achieves [INSERT: accuracy X%] on the held-out test set, demonstrating that edge AI inference is practical on the Raspberry Pi 4 with an inference latency of [INSERT: X ms] per frame — well within the requirements of near-real-time plant monitoring.

The majority-voting segmentation scheme reduced false positive disease alerts compared to whole-image classification, particularly for images with significant background or partially framed leaves. The image quality gate (Laplacian variance + luminance check) successfully rejected [INSERT: X% of frames] as unsuitable for inference in

outdoor lighting conditions, preventing spurious disease alerts from motion-blurred captures.

The hysteresis irrigation controller maintained target soil moisture with a mean steady-state error of [INSERT: plus or minus X%]. The GPS + compass navigation achieved waypoint arrival within [INSERT: X.X m] mean error, consistent with the plus or minus 2-5 m CEP specification of the NEO-7M module. A key limitation was GPS availability in dense canopy environments. Future work will integrate ultrasonic obstacle detection and wheel odometry.

IX. CONCLUSION

This paper presented AgroNex-Pi, a low-cost autonomous agricultural robot integrating soil moisture sensing, automated irrigation, scheduled grass trimming, GPS-guided navigation, edge AI plant disease detection, and cloud telemetry on the Raspberry Pi 4. The system demonstrated [INSERT KEY RESULTS] in real-world testing, establishing it as a viable platform for small-scale precision agriculture applications. The dual-model TFLite architecture enables adaptation to different crop types by swapping the model file and labels without any code changes. Future directions include obstacle avoidance, INT8 quantization for Edge TPU deployment, multi-plant GPS tracking, and OTA model updates.

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REFERENCES

[1] Food and Agriculture Organization (FAO), "The State of Food and Agriculture 2021," FAO, Rome, 2021.

[2] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in Proc. IEEE CVPR, 2018, pp. 4510-4520.

[3] C. W. Bac, E. J. van Henten, J. Hemming, and Y. Edan, "Harvesting Robots for High-value Crops," J. Field Robot., vol. 31, no. 6, pp. 888-911, 2014.

[4] S. P. Mohanty, D. P. Hughes, and M. Salathe, "Using Deep Learning for Image-Based Plant Disease Detection," Front. Plant Sci., vol. 7, p. 1419, 2016.

[5] A. Khanna and S. Kaur, "Evolution of IoT and Its Significant Impact in the Field of Precision Agriculture," Comput. Electron. Agric., vol. 157, pp. 218-231, 2019.

[6] L. Emmi et al., "New Trends in Robotics for Agriculture," ScientificWorldJournal, vol. 2014, Art. no. 404059, 2014.

[7] TensorFlow Lite, "TensorFlow Lite Guide," Google, 2023. [Online]. Available: https://www.tensorflow.org/lite

[8] Raspberry Pi Foundation, "Raspberry Pi 4 Model B Datasheet," 2019. [Online]. Available: https://www.raspberrypi.com

[9] Texas Instruments, "ADS1115 Datasheet: Ultra-Small, Low-Power, 16-Bit ADC," SBAS444D, 2009.

[10] Honeywell, "HMC5883L 3-Axis Digital Compass IC," Datasheet, 2013.